

# ECO387C - Fall 2024

## Discussion 5

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### 1 The problem

We begin by stating a standard stochastic endowment economy considered in class. Consumers solve

$$\max_{\{c_t, a_{t+1}\}_{t=0}^{\infty}} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \frac{c_t^{1-\frac{1}{\sigma}}}{1-\frac{1}{\sigma}}$$

subject to

$$c_t + \frac{a_{t+1}}{1+r} = y_t + a_t$$

and a borrowing constraint  $a_{t+1} \geq a^{\min}$  where  $a^{\min} \leq 0$ . The income process is stochastic and, to make it simple, we will assume that  $y_t$  is i.i.d. and discrete uniform on the interval  $[y^{\min}, y^{\max}]$ . The goal is to solve this model. In other words, we wish to express the choice variables as a function of state variables:

$$c_t = g_c(a_t, y_t), \quad a_{t+1} = g_a(a_t, y_t)$$

In the first discussion and problem set 1, we considered a deterministic version of this model. In those cases, we were able to obtain closed-form solutions for  $g_c$  and  $g_a$ . However, the model has now become complicated enough and does not yield such analytical solutions. Thus, we are going to solve this model *numerically* on the computer.

The state variables, which the agent takes as given, are  $y_t$  and  $a_t$ . The control variables are his consumption ( $c_t$ ) and saving ( $a_{t+1}$ ). Rewriting the agent's problem in the recursive form using the conventional notation, we obtain the following Bellman equation:

$$V(a, y) = \max_{a' \geq a^{\min}} \left\{ \frac{1}{1-\frac{1}{\sigma}} \left( y + a - \frac{a'}{1+r} \right)^{1-\frac{1}{\sigma}} + \beta \mathbb{E} [V(a', y')] \right\}$$

Notice that I have taken care of the borrowing constraint by only allowing values greater than  $a^{\min}$  in the maximization.

Let's expand the expectation on the right-hand side of the Bellman equation. If we let  $p(y')$  denote the probability that a particular  $y'$  materializes, then we can write

$$V(a, y) = \max_{a' \geq a^{\min}} \left\{ \frac{1}{1 - \frac{1}{\sigma}} \left( y + a - \frac{a'}{1+r} \right)^{1 - \frac{1}{\sigma}} + \beta \sum_{y'} p(y') V(a', y') \right\}$$

## 2 Conceptual numerical challenges

Numerically, there are a number of numerical challenges that we have to get a handle on:

1. *Function approximation:* Ultimately we will be interested in finding the solution to a dynamic programming problem. The solution to a dynamic program is a value function and a policy function. Both of these functions are mappings from the state space to the real numbers. For instance, in our case the savings choice will be captured by  $a' = g_a(a, y)$  and the associated value function is  $V(a, y)$ . In order to find these functions, we will have to be able to store them on the computer. One option is to view the value function as a matrix, or more generally, as a multidimensional array. In our case with 2 state variables, we could store the value function in a matrix of the following form. Discretize the state space of assets  $A = (a_1, a_2, \dots, a_{n_A})$  and do the same for income  $Y = (y_1, y_2, \dots, y_{n_Y})$ . Then we can view the value function as a matrix

$$V(a, y) = \begin{pmatrix} V(a_1, y_1) & V(a_1, y_2) & \cdots & V(a_1, y_{n_Y}) \\ V(a_2, y_1) & V(a_2, y_2) & \cdots & V(a_2, y_{n_Y}) \\ \vdots & \vdots & \ddots & \vdots \\ V(a_{n_A}, y_1) & V(a_{n_A}, y_2) & \cdots & V(a_{n_A}, y_{n_Y}) \end{pmatrix}.$$

This is called discretization and it is often the easiest and most robust option. However, we will see that it is not generally very efficient. Often, a better alternative is to specify a coarser grid and then to interpolate between those grid points. Linear interpolation would be one possibility.

2. *Optimization:* We will need to use numerical optimization in order to iterate on the value function. In the simplest case, in which the value function is discretized, we will simply choose the maximum from a vector. This is known as grid search. If the value function is not discretized, we will need to use numerical optimization tools.
3. *Numerical integration:* The problem we are considering involves uncertainty about the endowment in the next period. Thus, we need to evaluate expectations. In the case in which the value function is discretized, an integral is simply a weighted sum. If it is not, we will use numerical integration methods.

### 3 Value function iteration

Let's assume that the support of  $y$  is  $Y = \{y^H, y^L\}$  and that every period  $y$  is drawn from this set with probabilities  $(p, 1 - p)$ . Here is an outline of the algorithm.

1. Discretize the state space of  $a$  and  $y$ :

$$a, a' \in A = \{a_1, a_2, \dots, a_{n_a}\} \quad y \in Y = \{y^H, y^L\}$$

- If the array  $A$  is in ascending order,  $a_1$  will correspond to the borrowing limit  $a^{\min}$ . In other words, you can impose the borrowing limit by forming the gridpoint from  $a^{\min}$ .
2. Choose a starting value (initial guess) for the value function  $V_0(a, y)$ . For example,  $V_0(a, y) = 0 \quad \forall a, y \in A \times Y$

$$V_0(a, y) = \begin{pmatrix} V(a_1, y^H) & V(a_1, y^L) \\ V(a_2, y^H) & V(a_2, y^L) \\ \vdots & \vdots \\ V(a_{n_a}, y^H) & V(a_{n_a}, y^L) \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \\ \vdots & \vdots \\ 0 & 0 \end{pmatrix}$$

3. For a given value function  $V_j(a, y)$ , find  $V_{j+1}(a, y)$  by carrying out the following optimization

$$\begin{aligned} V_{j+1}(a, y) &= \max_{a' \in A} \left\{ \frac{1}{1 - \frac{1}{\sigma}} \left( y + a - \frac{a'}{1+r} \right)^{1 - \frac{1}{\sigma}} + \beta \sum_{y' \in Y} p(y') V_j(a', y') \right\} \\ &= \max_{a' \in A} \left\{ \frac{1}{1 - \frac{1}{\sigma}} \left( y + a - \frac{a'}{1+r} \right)^{1 - \frac{1}{\sigma}} + \beta \left( p \cdot V_j(a', y^H) + (1-p) \cdot V_j(a', y^L) \right) \right\}. \end{aligned}$$

- If for a given pair  $(a, y)$ , a choice  $\tilde{a}' \in A$  results in negative consumption  $c = y + a - \frac{\tilde{a}'}{1+r} < 0$ , we can simply set the associated utility to  $-\infty$ . This ensures that the max operator will not select  $\tilde{a}'$  as the optimal solution.
4. Repeat step 3 until the value function  $V_j(a, y)$  converges. Typically we use convergence criteria of the following form:  $\|V_{j+1}(a, y) - V_j(a, y)\| < tol$ , for some tolerance  $tol$ . Call this limit  $V(a, y)$ . If Blackwell's sufficient conditions for a contraction are satisfied, the above mapping will always converge.
  5. Find the policy associated with the limit  $V(a, y)$ :

$$g_a(a, y) = \arg \max_{a' \geq a^{\min}} \frac{1}{1 - \frac{1}{\sigma}} \left( y + a - \frac{a'}{1+r} \right)^{1 - \frac{1}{\sigma}} + \beta \sum_{y'} p(y') V(g_a(a, y), y').$$

- Then, you can also back out the optimal consumption  $c_t^* = g_c(a, y)$  using the budget constraint.